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IF3E Carpooling website project report



Introduction :

The IF3E UE was designed to teach students the basics of the relational databses and this mid-semester project was here to help students gather all the stuff they learnt during this half-semester in one big project.

Working on this project, I had to design the architecture of a database, I had to write the SQL resquests needed to gather the information and I had to write the PHP needed in order to put together the power of the relational databases with a web browser.

All these skills were completly new for me so I had to work two weeks on this website in order to use all the skills I’ve learnt during the lectures.

In this report, you will see the different sides of the project such as the relational model used or the implemented features.

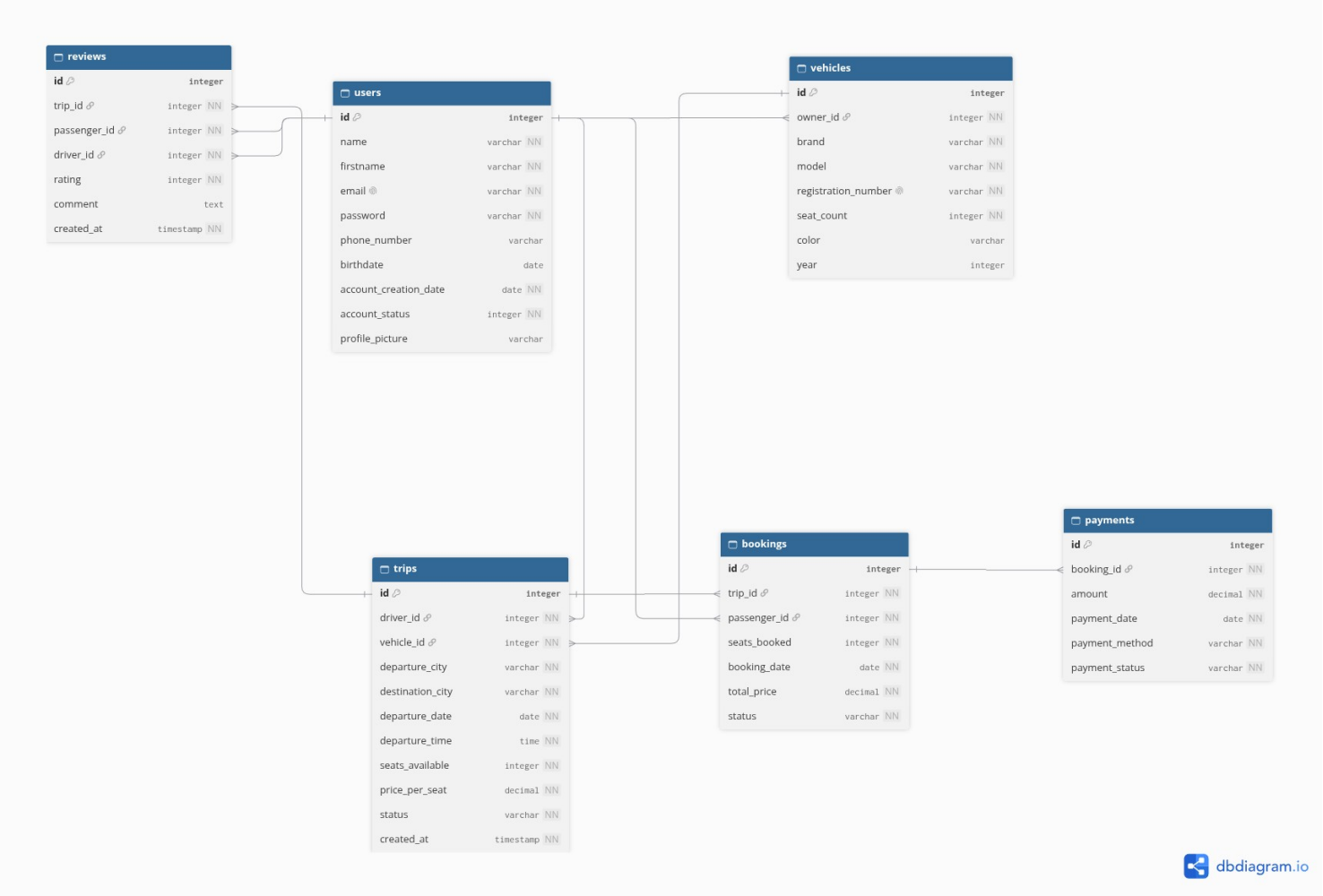
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The entity-relationship diagram and the relational model

Here is the entity-relationship diagram used to represent the different relations between the tables in the project. This diagram has been created with dbdiagram.io :



Since the picture is quite small, you'll have to zoom in order to see the relational model correctly. However you can access it in this doc/ directory with « ERD.png ».

Basically, the table named users will contain all the infromation required about the users and all the information needed to login. It contains a primary key named id.

Some users may be drivers, that's the reason why the table users is connected with the table vehicles which contains the information required about the vehicles posessed by the users which will be used for carpooling. The table contains a primary key named id and a foreign key connected to users named owner_id.

Next, there is the trips table. When the users want to drive somewhere, the may decide to invite other users of the website to join them. They will need to create a new trip specifiying which vehicle will be used for the trip, and other information about this one (What is the departure city and the destination city ? When will it be ? …). The trips table contains a primary key named id and

two foreigns keys ; the first one connects the table with the users table, it’s the driver unique id named driver_id. There is also a foreign key named vehicle_id connected to the vehicles table who links the trips table with the vehicle table.

Next, an interested passenger will probably book the trip. That’s the reason why the bookings table is here. This table will contain the information about the trip selected and the passenger who wanted to book the trip. As usual, there is always the id primary key, but there is also a foreign key connecting the booking with the users table named passenger_id and there is also another foreign key linking the trips table with the booking table ; trip_id.

The passenger will need to pay for its booking and this payment will be recorded in the payments table with some information about the payment such as the method used. The unique id is the primary key of the payments table but there is also the foreign key booking_id which connects the tables payments and bookings.

Finally, once the trip is over, a passenger may eventually leave a comment on the website about the driver, which will be saved in the reviews table alongside with the rating and the ids of the passenger and the driver. There is one primary key id, but there are here three distincts foreign keys ; two linking to the users table passenger_id and driver_id that respectfully reference the passengers and the drivers ids and trip_id which is connected to the trips table.

Explanation of 3NF normalization

During the lectures of this semester, we've learnt that there is some rules that have to be respected in order to get a working database.

The first rule that has to be respected is named 1NF. 1NF states that only one information should belong to a cell. For example, it is a bad habit to write a user database that contains a cell with the name of the user and another cell that contains his phone number and his adress at the same time. All the information should be separated in their proper column. That's the reason why in the whole database, there isn't any table where there is more than one information belonging to the same cell, names, firstnames, email adress, phone numbers but also trip ids, payment method ; all those information are separated in their proper columns. That's the reason why 1NF is respected.

There is another rule that we have learnt which is 2NF. 2NF states that all non-key attributes must depend on the entire primary key. This problem is easily solved in the project by adding to all tables a unique id that permits to identify in a unique way all entries.

There is finally a last rule to respect : 3NF. 3NF does not accept transitive information. For instance, in the trips table, there is a foreign key that references the id of the vehicle used for the trip. At this place there is only a unique id that references the vehicle, no other information about the vehicle such as its brand or its colour ; all these informations stay only in the vehicles table and are therefore not repeated in the trips table, they can be accessed only with the unique vehicle_id. This helps reducing errors when editing the database where data may become irrelevant such as the vehicle number 5 that has been used in two different trips but was blue the first time and green the second. When 3NF is respected, those accidents could not happen.

Implemented features

This project includes the basic required features asked in order to respect the instructions.

We can basically use the website for creating new accounts by using the registering option and log in to them. And as previously seen in TDS, the passwords are hashed before being saved in the database.

As stated in the instructions, the project offers the possibility for a user to register one of his own cars. The user has to give some information such as the coulour, the year, the brand or the registration number of the car. Once this is done the car is registered in the appropriate table.

Now that the user has registered his own car, it is possible for him to publish a new trip. The user will need to precise the date and hour of the trip with the car used and the number of seats available. Once the trip is registered, it will possibly be accessible to the other users.

A user can simply search for interesting trips on the website. He may use the tools provided by the website to filter the different trips and book some of them if he is interested by one of them.

Once a user has booked a trip, he can pay for the trip, by clicking on a link on his dashboard. He will then be redirected to a page where he can leave his credit card information. However the interface is not true and as long as the information respect the desired format, the card will be accepted and the website will consider that the trip is paid.

A driver may also cancel one of his trips or mark them as completed. Once a trip is completed, a user may leave a review which will be accessible to the driver of the trip.

There are also basic admin tools which permit to the admins to give the admins privileges to other basic users and see basic statistics of the website.

Moreover, there are also a few optional features that were not required in the project but are still present in the website such as an international telephone format helper tool in the registering page

The registering system ensures that the fields are properly completed (correct email format, correct phone number format).

However, there is some features that couldn’t be implemented such as a more powerful admin system and other edition and delete features.

Conclusion

The basic idea of the UE IF3E was to teach how to use relational databases and how to interact with them. Even though there are a lot of possible uses in a programmer life, the exercises still stayed a bit abstract and were not totally enjoyable since the purpose of the TD’s exercise was only to help us practicing what we just learnt during lectures.

On the other hand, this project taught us how to handle relational databases, SQL and PHP in a completly other way ; I had to work in a specific environment with a specific context and creating a full website.

To be honest, I found this project very difficult, not only because I never saw how to use PHP and how to connect it with SQL before, but because of the small amount of time given too. I only had two weeks to do the project, which was quite stressful I must admit but also very rewarding too.

I want to conclude this report by saying that I am quite proud of me even if I had to use a lot of external resources such as stack overflow, ai tools or w3schools in order to succed. I am sure there is a huge amount of optimization and other cool new features I could add to the website, I could for instance add a ban-system for admins or a chat system for the users, but time lacked so I decided to go only right to the goal and deliver the best website I could possibly submit in two weeks.

I hope that you will enjoy my brand new website !